



FARAH TSABITAH AFLAH

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A fresh graduate of Informatics Engineering at Universitas Riau with 3 full years of scholarship and experience in AI-based chatbot development, mobile development, and web development. Interested in artificial intelligence, data analytics, natural language processing, and user-centered digital solutions. Eager to keep learning, take on new challenges, and contribute to impactful technology projects.

EDUCATION

Universitas Riau

2022 – 2026

- Major in Informatics Engineering | GPA: **3.82**
- Recipient of the Riau Provincial Government Scholarship

SKILLS

- **Data & ML** : Python | Pandas | NumPy | Scikit-learn | SQL | RAG | NLP | LLM
- **Development** : Kotlin | Java | PHP | JavaScript | Laravel | Android | MySQL
- **Software** : Microsoft Office | Visual Studio Code | GitHub | Google Colab | Streamlit
- **Soft Skill** : Analytical thinking | problem solving | attention to detail | communication | team collaboration | adaptability
- **Language** : Bahasa Indonesia (native) | English (advanced, TOEFL Score: 557)

EXPERIENCE

Software Engineering Laboratory | Riau

Feb – Jun 2026

Teaching Assistant

- Prepared and delivered practical modules for the Digital Systems course using Logic Gate Simulator.
- Mentored students during laboratory sessions covering logic gates, combinational circuits, flip-flops, half/full adders, multiplexers/demultiplexers, and seven-segment displays.
- Conducted laboratory consultations, practical examinations, and evaluated technical reports, providing feedback to strengthen students' analytical and documentation skills.

Dinas Komunikasi Informatika dan Statistik Provinsi Riau | Riau

Dec 2025 – Jan 2026

Software Engineer Intern — Aplikasi Teknologi Informasi dan Komunikasi

- Performed Quality Assurance (QA) and user acceptance testing (UAT) for web applications developed by the division, identifying functional issues and verifying system compliance with user requirements before deployment.
- Developed SIPENA, a Laravel-based administrative management system for regional government agencies (OPD), implementing backend CRUD functionality, server-side DataTables, and a 20-table MySQL relational database to streamline administrative submission workflows.

RESEARCH

Academic Chatbot Performance Comparison — RAG vs Rule-Based

Dec 2025 - May 2026

Bachelor’s Thesis

- Collected, preprocessed, and transformed academic documents using Pandas and NumPy, cleaning and structuring data for Rule-Based and RAG pipelines.
- Developed a hybrid RAG pipeline using LangChain, ChromaDB, BM25, vector retrieval, and SeaLLM for context-aware question answering.
- Evaluated chatbot performance using Accuracy, Precision, Recall, F1-Score, Answer Relevancy, BERTScore, and Response Time, with RAG achieving 94% Accuracy, 97% F1-Score, and 95% Answer Relevancy.

Bangkit Academy led by Google, Tokopedia, Gojek, & Traveloka

Sep 2024 - Jan 2025

Capstone Project – DiaBite: AI-Powered Diabetes Prediction & Recommendation

- Developed an Android application integrating a cloud-hosted AI/ML prediction model deployed on Google Cloud Platform via Retrofit.
- Implemented secure user authentication with OTP verification, local data persistence using Room Database, and analysis history management.
- Integrated Gemini API to provide AI-generated diabetes education and fetched curated diabetes-related articles to enhance user health awareness.
- Collaborated in a multidisciplinary capstone team to develop and deploy cloud-based AI services using USD 100 in Google Cloud Platform credits provided by Bangkit Academy.

PROJECT

Mental Health Risk Analysis using Clustering

- Analyzed a Kaggle dataset of 2,000 individuals through preprocessing, feature selection, and clustering analysis.
- Compared K-Means and Agglomerative Clustering using Silhouette Coefficient, Davies–Bouldin Index, and execution time.
- Identified three mental health risk segments to support data-driven mental health interventions.

CRUD Portfolio Website

- Built a responsive personal portfolio website with secure authentication and an administrator dashboard using Laravel and Filament.
- Implemented complete CRUD functionality to manage website content dynamically without modifying source code.
- Designed responsive layouts to provide an optimal user experience across desktop and mobile devices.

CERTIFICATIONS

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|-----------------------------------|--------------------------|---|--------------------------|
| • Mobile Development (Android) | (Link) | • Oracle Database Design | (Link) |
| • BNSP — Junior Office Operator | (Link) | • Oracle Database Programming | (Link) |
| • BNSP — Online Sales Officer | (Link) | • Top 50 teams at Bangkit Product Track | (Link) |
| • FGA — AI Beginner | (Link) | Capstone Project | |
| • FGA — AI Intermediate: AI Track | (Link) | • Bangkit Distinction Graduate | (Link) |